

Experience with Google Colab for running Modelica FMU in notebooks and no installation

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Google Colab provide the user for free with a virtual machine with Ubuntu, Python and their branched version of Jupyter notebook. This can be used for demonstration and teaching using simulation of Modelica FMUs with PyFMI [1] or FMPy [2]. In this way simulation also becomes available on tablets or even smartphones. The work is inspired by the tutorial [3].

I have for a couple of years maintained a Github website where basic examples of bio-processes [4] can be run from the web-browser using Google Colab and no installation needed. The examples are documented as notebooks and can be freely continued by the user through command-line interaction using a simplified concise set of Python functions and workspace dictionaries [5] that provide a common meta level to PyFMI and FMPy interaction.

In this talk I share some of my experience. Essentially, I think the technology and tools are mature enough. I would like to further facilitate “Github-creativity” on the level of notebooks. The tools presented here hopefully might be useful in other application areas where simulation is more commonly used compared to biotechnology.

[1] Winther C et al <https://github.com/modelon-community/PyFMI>

[2] Sommer T et al <https://github.com/CATIA-Systems/FMPy>

[3] Sommer T et al <https://github.com/t-sommer/fmi-cloud-tutorial>

[4] Axelsson, J. P., “*Design ideas behind Bioprocess Library for Modelica*”, Modelica 15, Aachen, Oct 9-11, 2023.

[5] Axelsson, J. P., “*Design aspects of FMU-explore a Python module to complement PyFMI*”, Linköping, 2022.